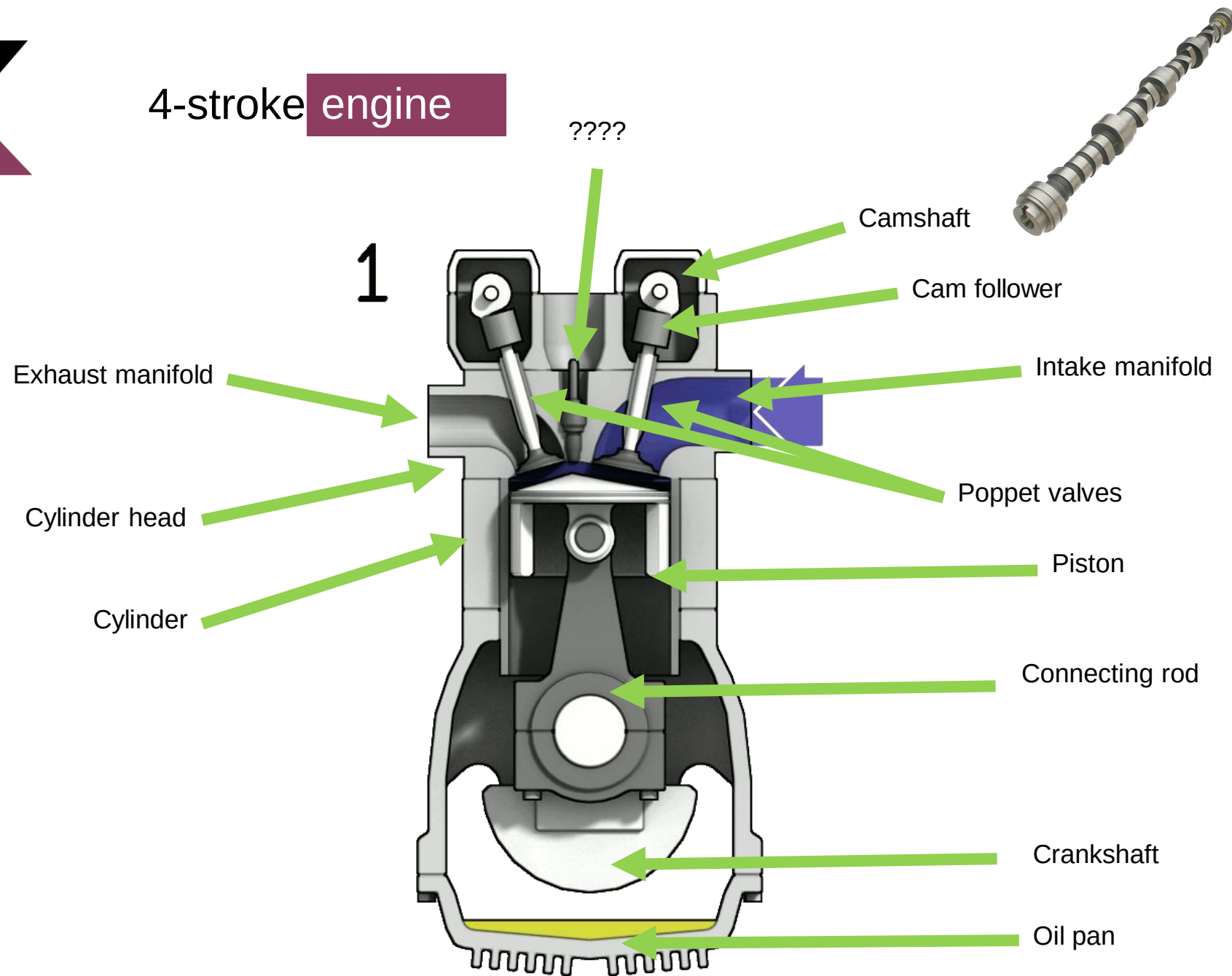


MOTORI A
COMBUSTIONE
INTERNA



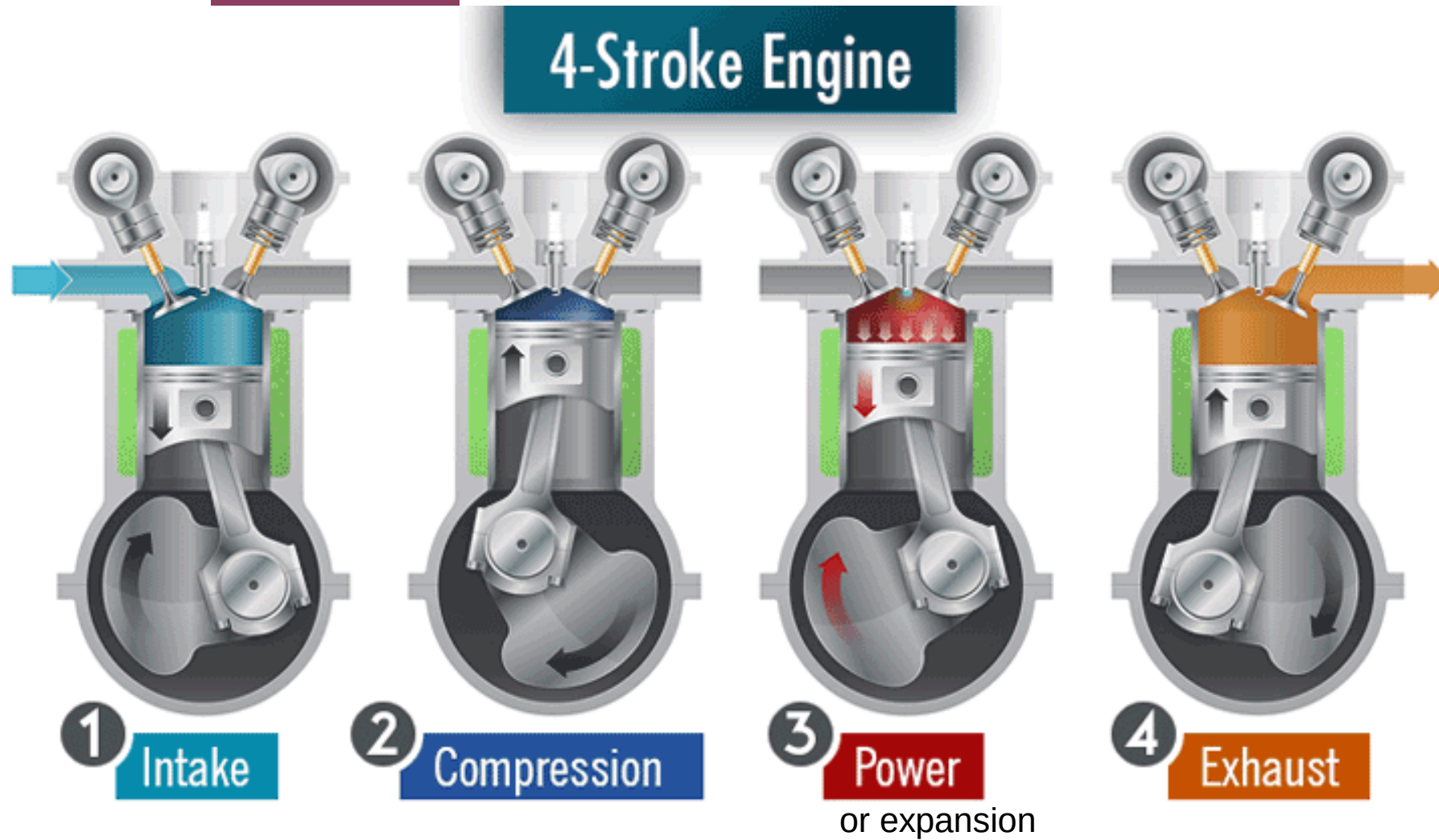
4-stroke engine





4-stroke
engines

Operation





4-stroke engines – Relevant parameters



- Displacement

$$displacement = s * \frac{\pi B^2}{4} * N_c$$

- Compression ratio

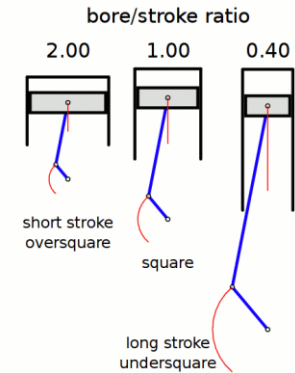
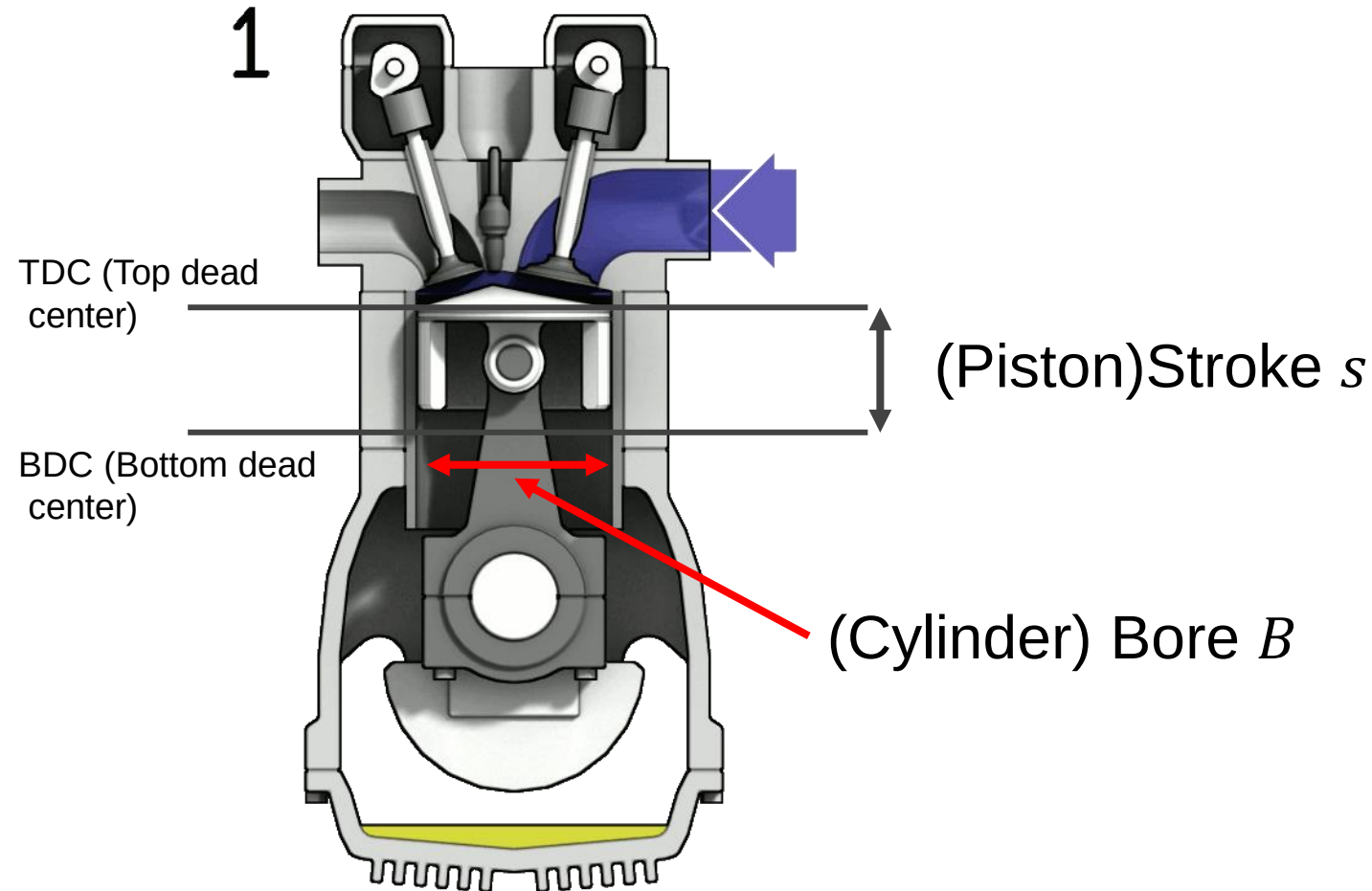
$$CR = \frac{volume\ at\ BDC}{volume\ at\ TDC}$$

- Stroke ratio

$$Stroke\ ratio = \frac{B}{s}$$

- Air-fuel ratio

$$AFR = \frac{m_{air}}{m_{fuel}} = \frac{\dot{m}_{air}}{\dot{m}_{fuel}}$$

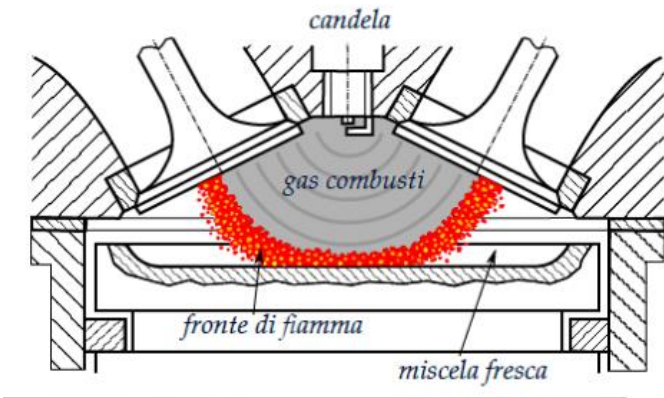




aka 'gasoline'

Combustion is *premixed* and is triggered by one (or two) **spark plug(s)**

1



Air-Fuel mixture is **stoichiometric** and constant (*although lately this is not always true anymore*)

Torque (hence power) is regulated **by modulating the airflow**

Roughly speaking the engine power can be calculated as:

$$P = \eta(T_e, rpm) * \dot{m}_{fuel} * H_i = \eta(T_e, rpm) * \text{AFR} * \dot{m}_{air} * H_i$$

CONSTANT FOR A SI ENGINE!!!

kineton

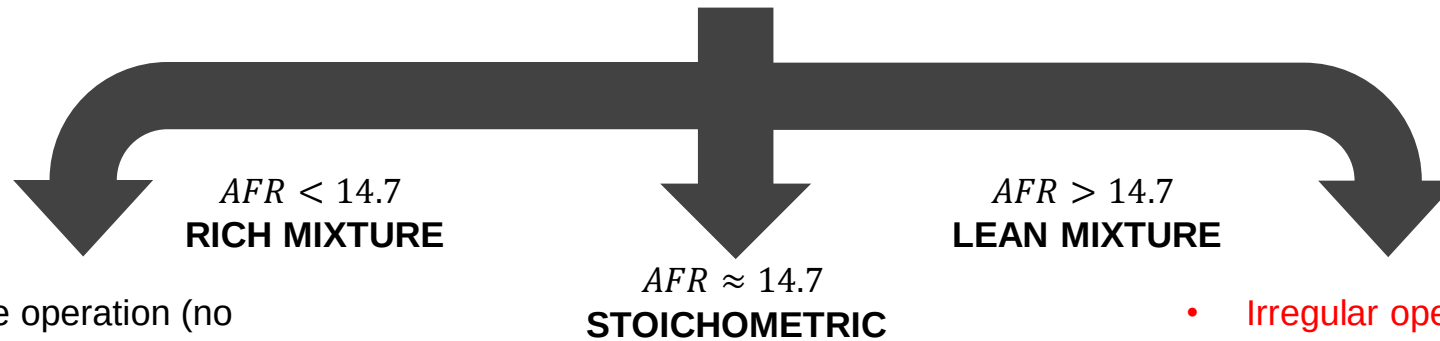


Spark Ignition Engines (SI)

Stoichiometric ratio

Why having a stoichiometric mixture is so important?

$$AFR = \frac{m_{air}}{m_{fuel}} = \frac{\dot{m}_{air}}{\dot{m}_{fuel}} \approx 14.7 \text{ (gasoline)}$$



- Apparently normal engine operation (no damage risk) if mixture is not excessively rich.
- Partial fuel oxidation produces HC, CO (pollutants), soot.
- Excessive fuel consumption (more fuel than necessary is used)

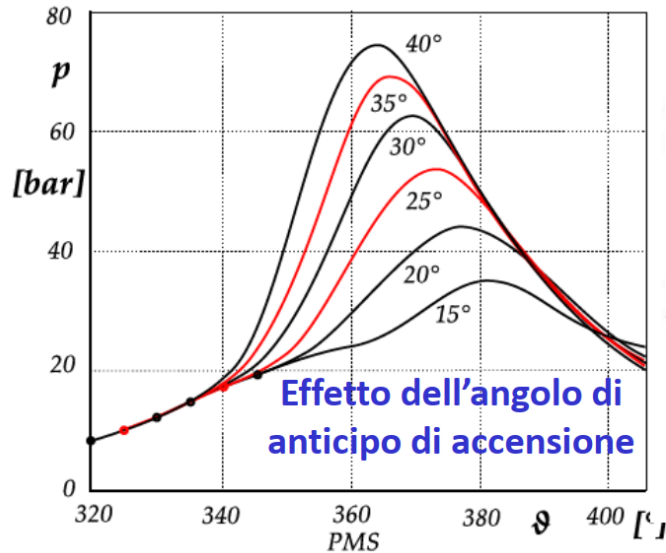
- Ideal engine operation
- Maximum fuel oxidation ideally yields CO_2 and H_2O only

- Irregular operation (misfire)
- Slower flame front may cause overheating.
- Overheating may cause engine knock and pre-ignition.
- Increased mechanical and thermal stresses
- In some cases, the efficiency may be increased by adopting a lean mixture



Spark Ignition Engines (SI)

Combustion

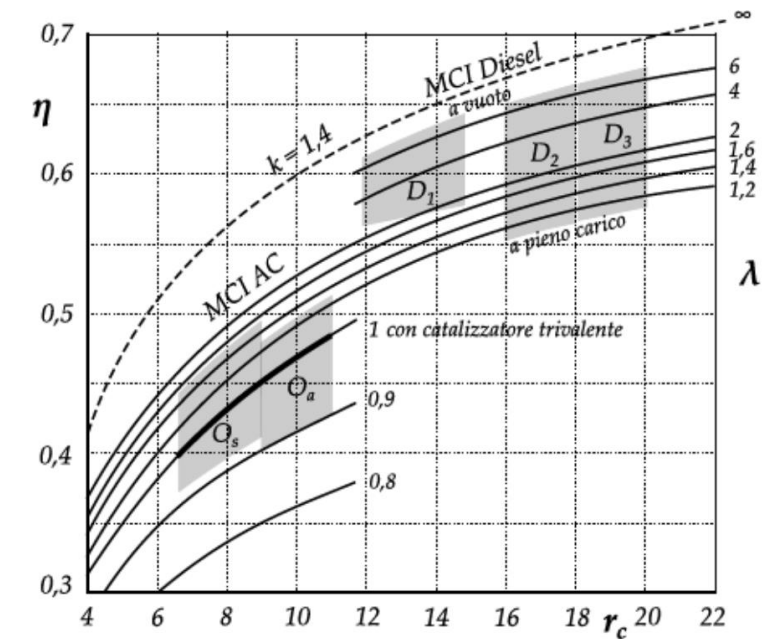


Spark advance (**RESPECT TO TDC**):

- Combustion does not start immediately after spark plug ignition. **ADVANCE**
- An higher spark advance increases the peak pressure (and temperatures), thus **power and efficiency (typical 15°/25° before TDC)**.
- Increases engine mechanical and thermal stresses and might lead to **engine knock** unless **higher OCTANE NUMBER** (typical 95 ÷ 100 for gasoline, up to >120 for methane) fuels are used.

Influence of the compression ratio:

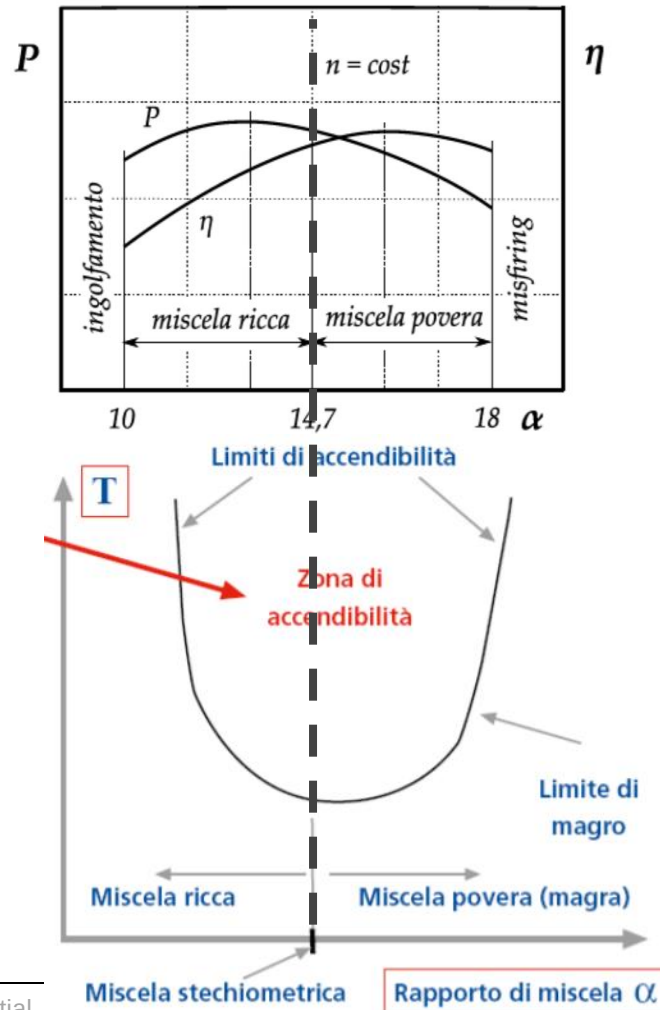
- **Higher compression ratios lead to higher efficiency**
- Increases engine mechanical and thermal stresses and might lead to **engine knock and self ignition** unless **higher OCTANE NUMBER** fuels are used.
- Reducing the spark advance may be necessary in order to avoid the aforementioned phenomena (thus reducing efficiency)
- Turbocharged engines generally use lower compression ratios (thus reducing efficiency).





Spark Ignition Engines (SI)

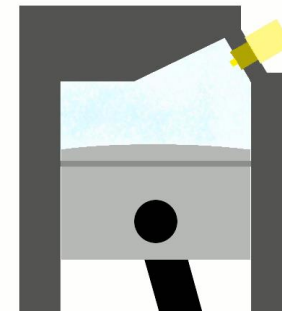
Misfire, Knock and Pre-ignition



Misfire: the fuel is **too much diluted** in the intake air. Combustion cannot occur. Unburned fuel is exhausted (pollutants).

Choke: the fuel is **too little diluted** in the intake air. Combustion cannot occur. Unburned fuel is exhausted (pollutants).

Knock: the combustion is triggered by some other hot spot (e.g. the exhaust valve) in the combustion chamber **after** the start of combustion by the spark plug (the flame fronts collide).



Pre-ignition: the combustion is triggered by some other hot spot (e.g. the exhaust valve) in the combustion chamber **before** the start of combustion by the spark plug.

May be caused by overheating or by a too high compression ratio (the mixture overheats by effect of the compression before the spark ignition).

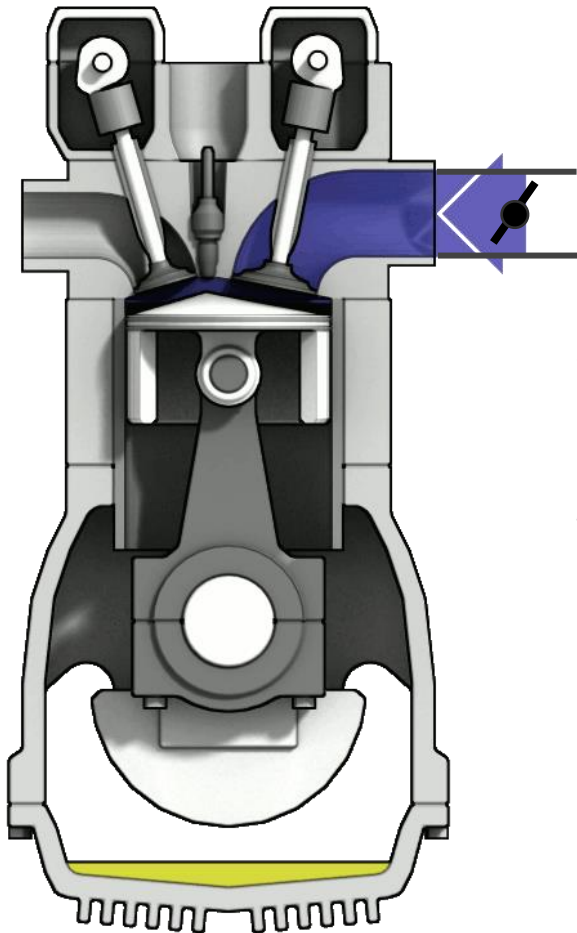


Spark Ignition Engines (SI)

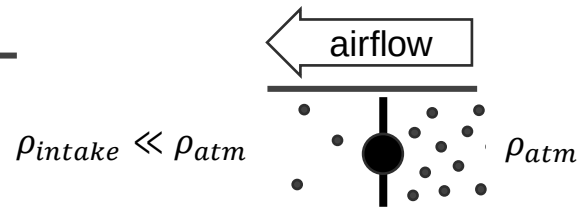
Control

How to modulate the air flow (i.e. the torque)?

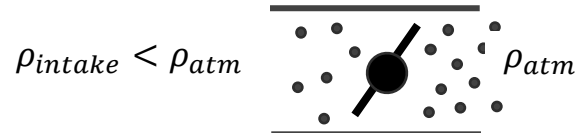
The throttle body “chokes” the intake flow thus reducing the intake air mass flow at constant volumetric flow.



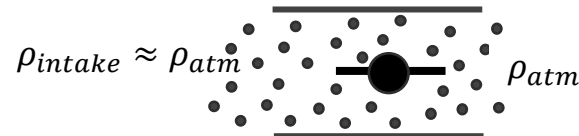
$$\dot{m}_{air} = \rho_{intake} \dot{V} = \rho_{intake} \cdot \frac{rpm}{2 \left[\frac{revs}{intake} \right] \cdot 60 \left[\frac{s}{min} \right]}$$



Closed throttle (e.g. idle) – air density downhill the throttle body is at minimum.



Medium throttle – the air density downhill the throttle body is higher but still lower than the atmospheric pressure



Full throttle – the air density downhill the throttle body is approximately the same as the atmospheric pressure

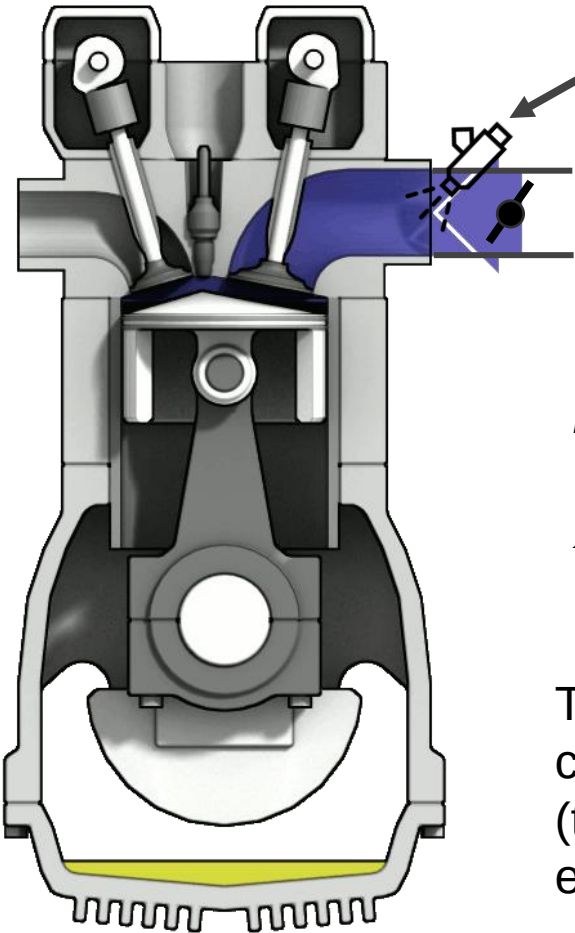
The throttle opening is commanded by the accelerator pedal



Spark Ignition Engines (SI)

Fuel flow control 1/3

1



Fuel is injected into the combustion chamber through a **fuel injector** located in the intake manifold downhill the throttle body (port injection), operating at low pressure (3-5bar).

Since the fuel is volatile (gasoline) or gaseous (LPG, methane), it is dissolved into the intake flow

This is not the only solution! Direct injection can be adopted as well

How much fuel is injected?

$$AFR = \frac{m_{air}}{m_{fuel}} = \frac{\dot{m}_{air}}{\dot{m}_{fuel}} \approx 14.7 \text{ (gasoline)}$$

AFR shall be constant and the closest possible to the stoichiometric ratio!

The **exact** air mass flow should be known **a priori** in order to be close enough to the stoichiometric AFR. That is nearly impossible (transients, fluid dynamics and thermal effects, fuel vaporization etc etc).

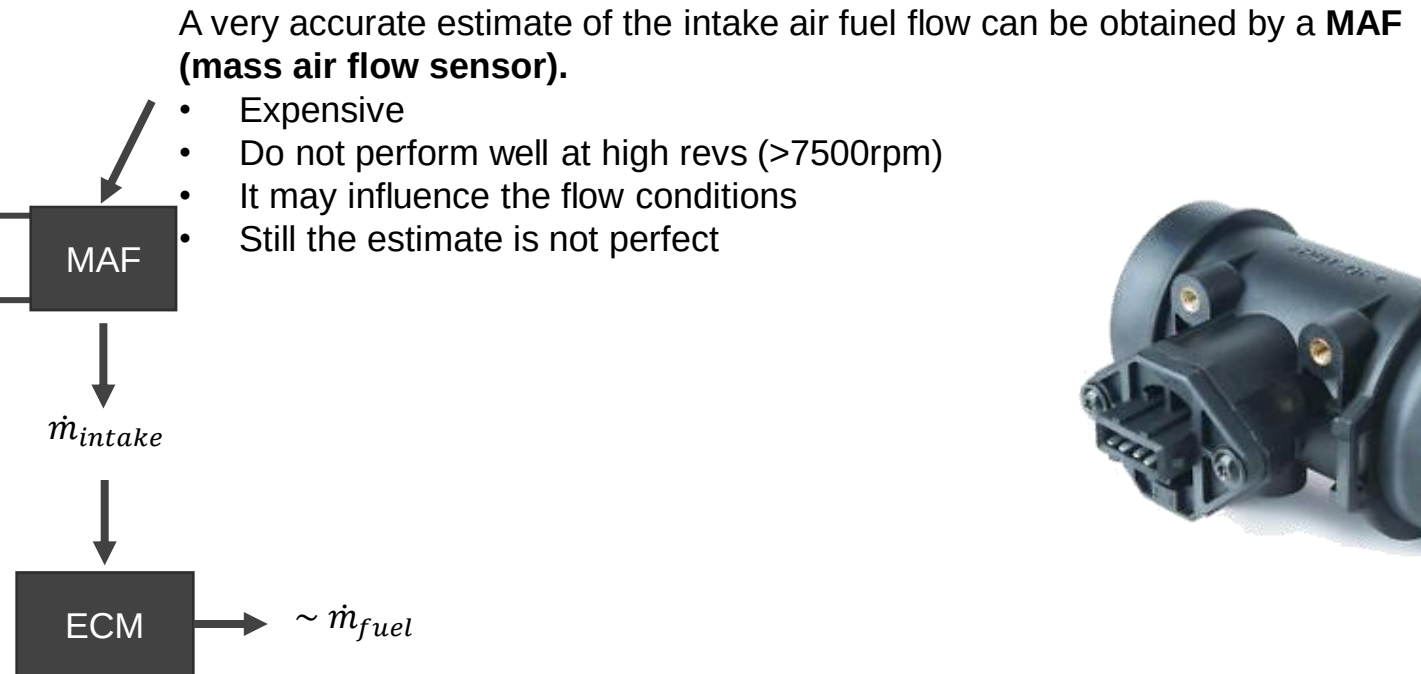
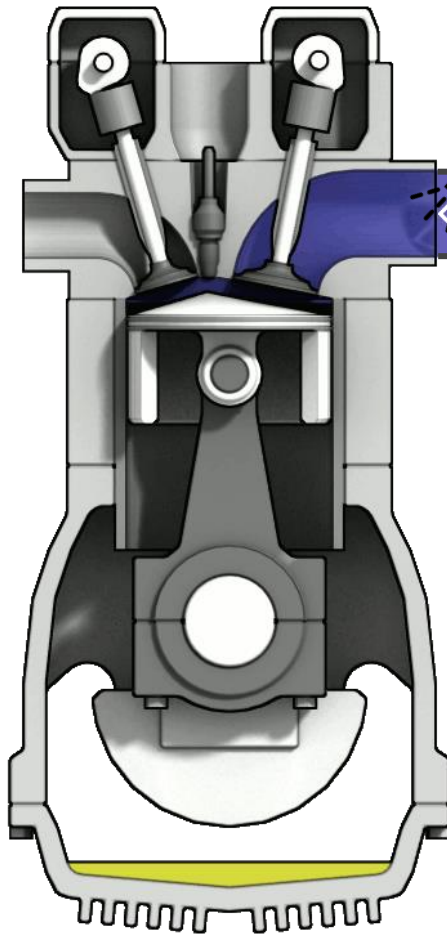




Spark Ignition Engines (SI)

Fuel flow control 2/3

1



Some feedback correction is needed in order to dynamically adjust the quantity of injected fuel.

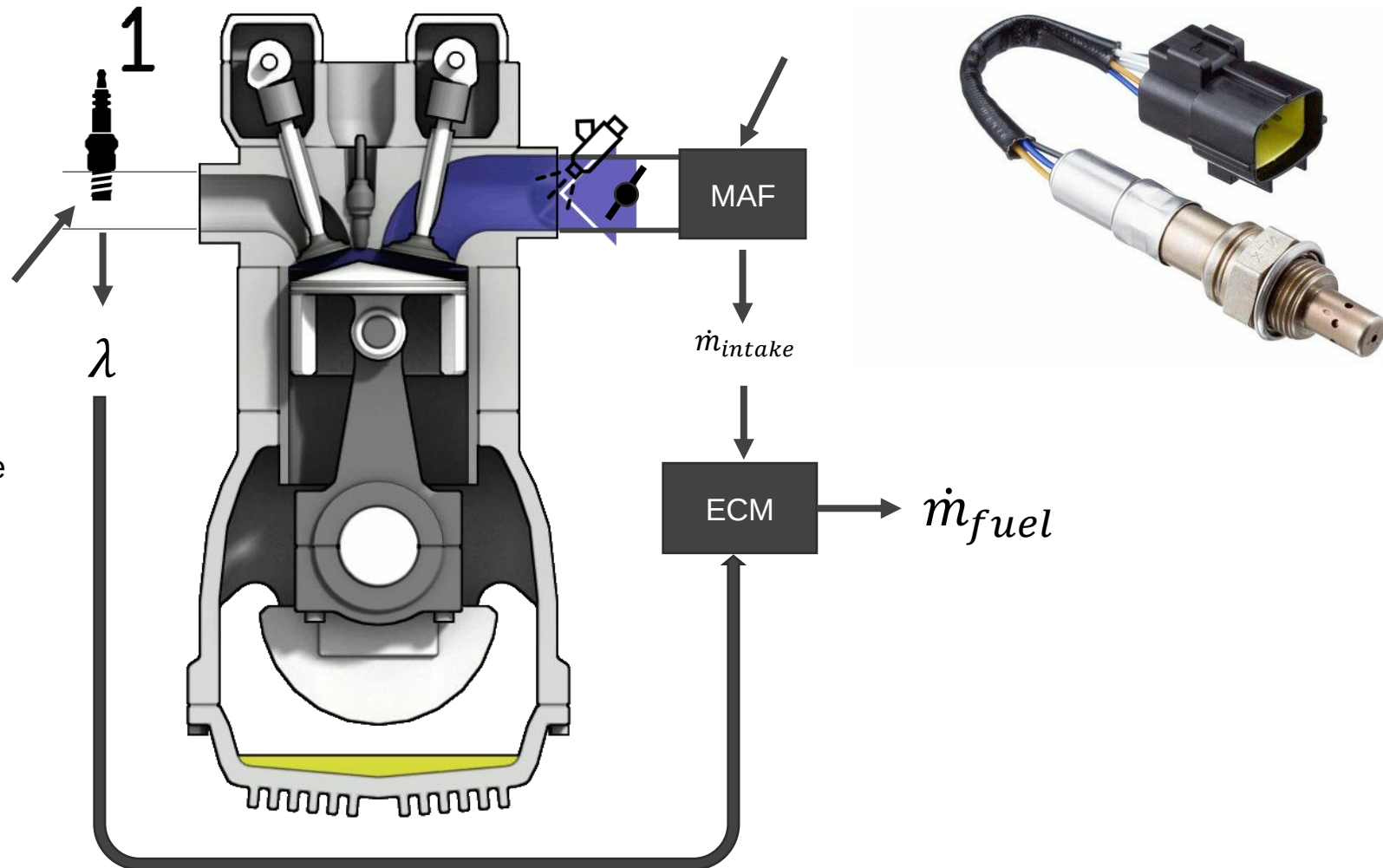


Spark Ignition Engines (SI)

Fuel flow control 3/3

The additional feedback correction is calculated by using information from the **lambda sond** located in the exhaust manifold.

The lambda sond detects the presence of excess oxygen at the exhaust (lean mixture) and provides this datum to the ECM





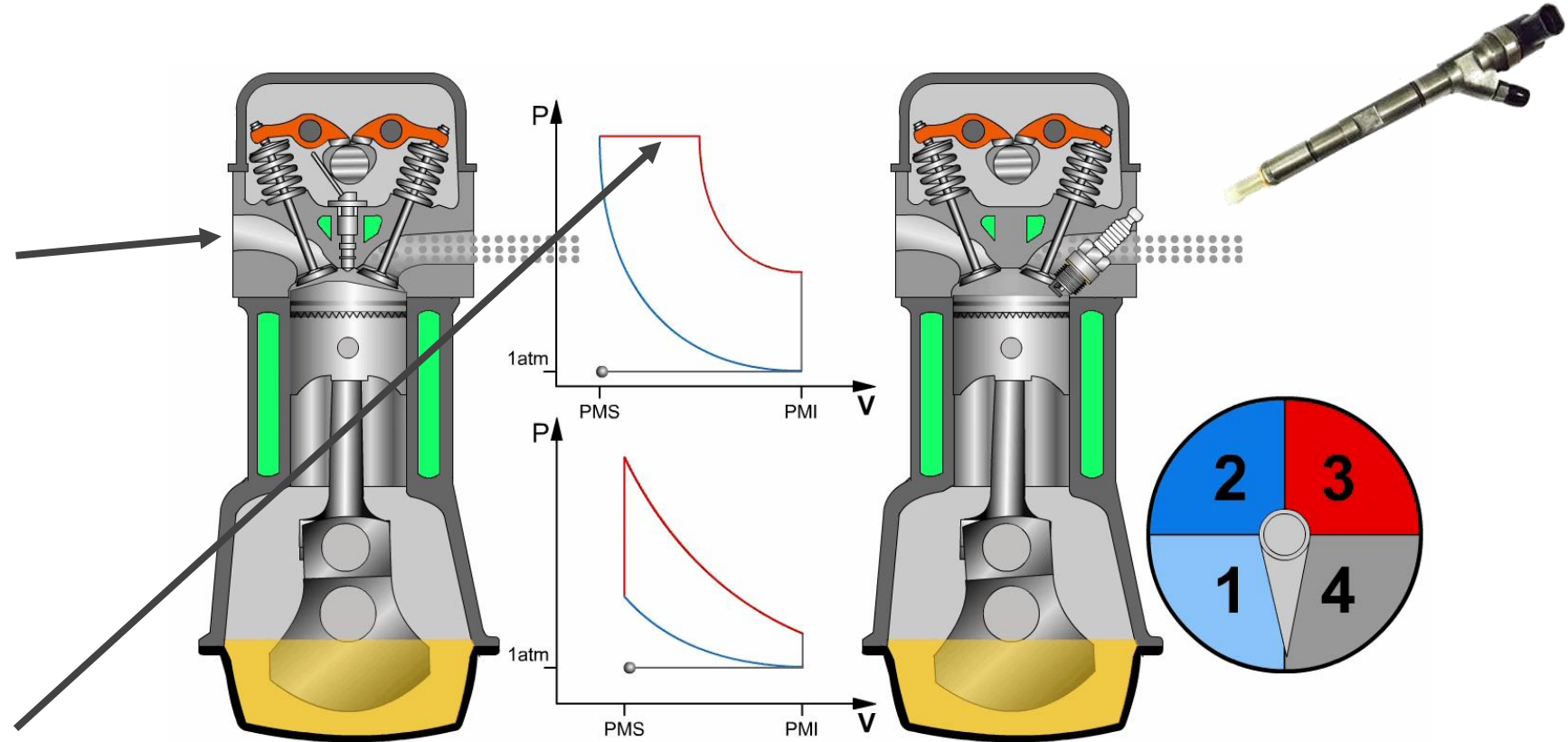
Diesel vs SI A quick comparison

The **intake is free**, there is no throttle body. The air flow mainly depends on engine speed only (rpm).

As a consequence, diesel engines **ALWAYS** operate with **excess air**.

The **combustion occurs along the expansion stroke** (slower than SI engines)

There is no spark plug, the combustion starts spontaneously for the high pressures and temperatures at the end of the compression stroke (**typical end of compression pressures are around 60bars 120-130bar, 2x of SI engines**).



In order to attain the higher pressure and temperatures, the compression ratio is higher than SI engines (15 ÷ 20)

Diesel fuel is not very volatile. It is injected at extremely high pressures (up to 2000bars) in order to form small droplets



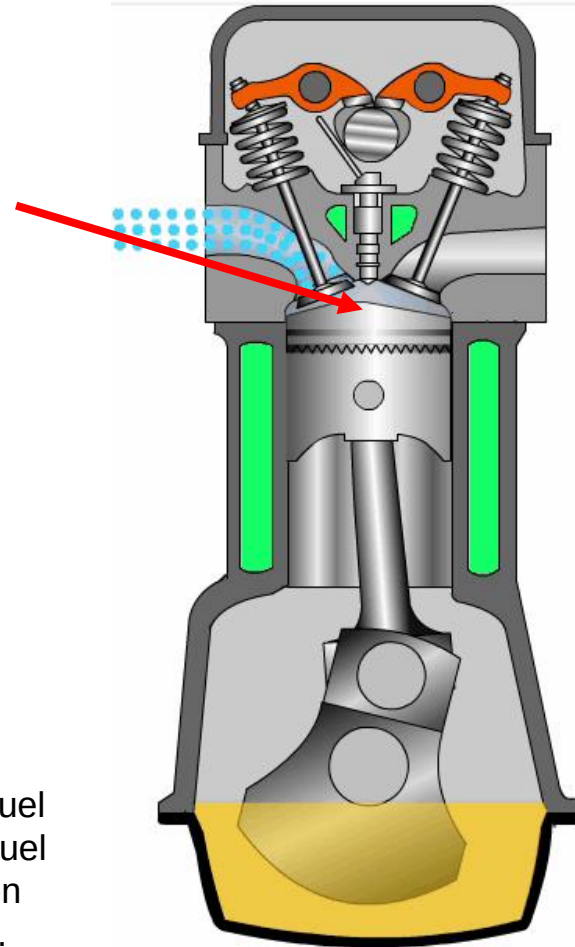
Diesel engines

Main Characteristics

The mixture is dishomogeneous within the combustion chamber: the **AFR varies from 0 in fuel droplets to infinity** in areas where air only is instantaneously present.

Fuel is injected when the pressure and temperature conditions to trigger combustion are already present, the combustion starts after a **temperature dependent ignition delay** after injection.

The shorter the ignition delay, the better the fuel quality. **The CETANE NUMBER** defines the fuel quality (typically 51 ÷ 60). With shorter ignition delay, higher engine speeds can be obtained.



The combustion mechanism is **mainly diffusive rather than premixed** (as in SI engines)

The **air mass** in the combustion chamber is approximately the same (differently from SI engines where the throttle modulates the density). **Torque and thus power regulation is obtained by directly varying the quantity of injected fuel.**

The absence of throttle (less pumping losses) and higher compression ratio, allow higher efficiencies compared to SI engines.



Diesel engines

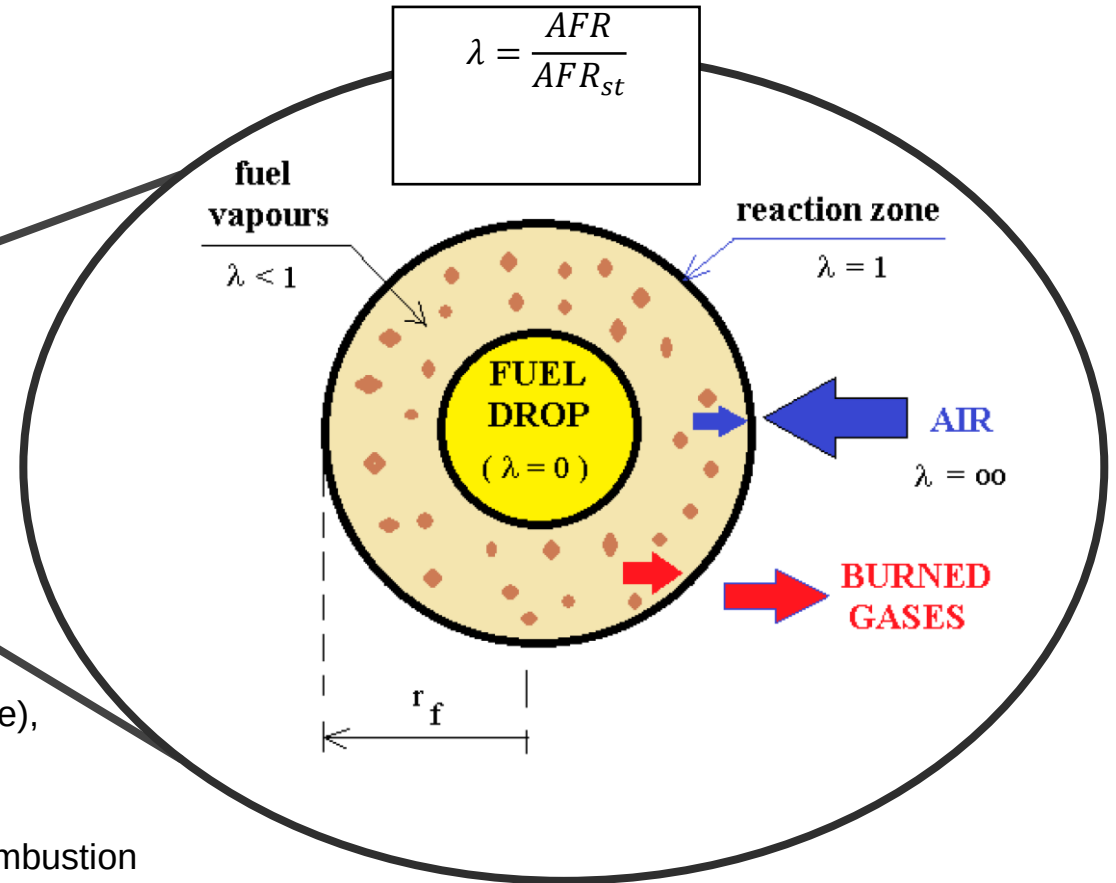
Combustion 1/2



As combustion progresses, the fuel drop evaporates (reduces its size), producing more fuel vapours.

If the oxygen concentration at the reaction zone gets too low, the combustion may stop (e.g. for poor jet mixing, low turbulence)→A fuel droplet surface is partially oxidized because of the high temperature (carbonization), **producing soot**.

Soot is also produced when the temperature is too high (the fuel drop surface is carbonized before it vaporizes completely).

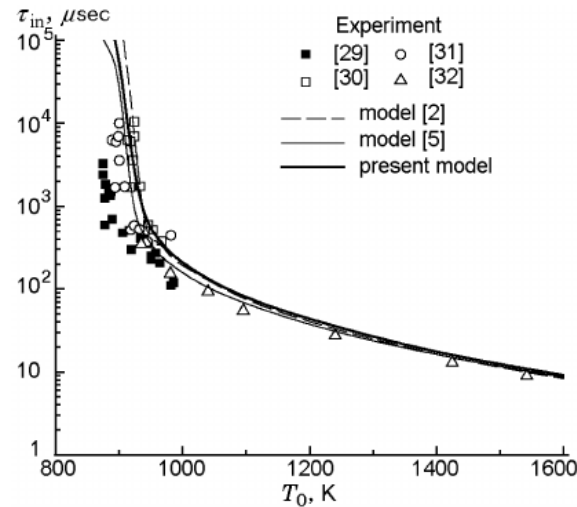


The smaller the fuel droplets are, the easier it gets for it to vaporize completely, thus limiting soot formation.



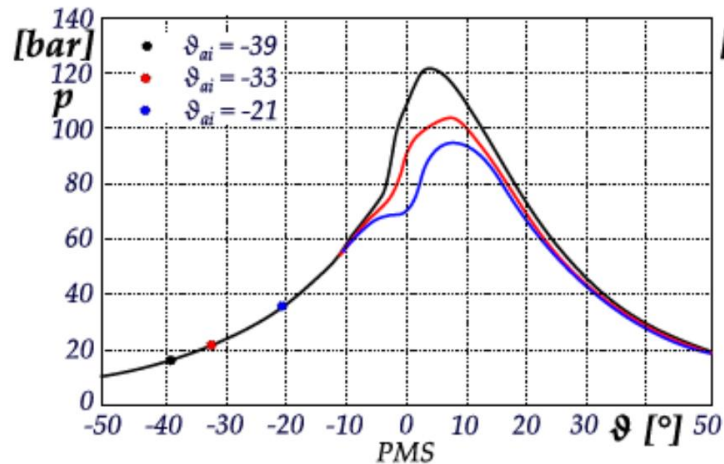
Diesel engines

Combustion 2/2

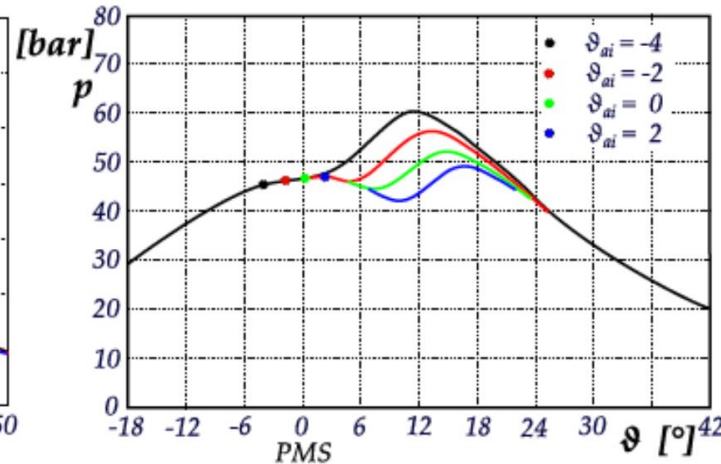


The ignition delay is extremely sensitive with temperature, 200K temperature difference may increase the ignition delay by 20 times (logarithmic time scale in figure).

Glow plugs are used for cold start in diesel engines, in order to create a "hot spot" where combustion can start more easily



(a) Iniezione anticipata



(b) Iniezione ritardata

Analogously to spark advance for SI engines, **increasing the injection advance** in Diesel engines increases peak pressure and temperature (thus efficiency), **but mechanical and thermal stresses are increased (and engine noise)**. Same considerations as SI engines apply.

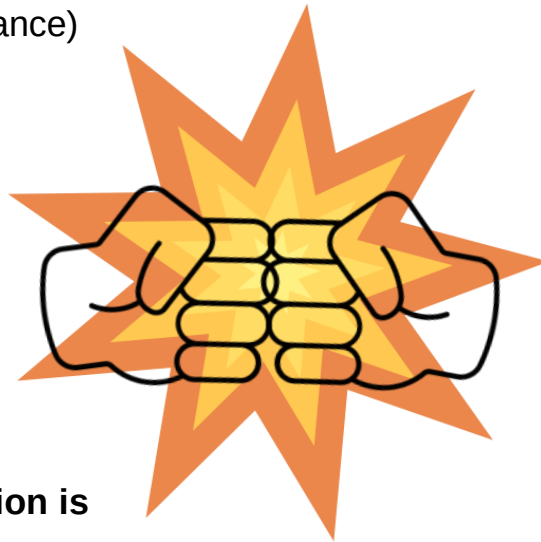


Soot production

- Too little oxygen (locally or globally)
- Poor air-fuel mixing (low turbulence)
- Poor vaporization (droplets too large)
- Temperature too low (too low injection advance)
- In some soot can be produced even when temperature is too high

Observations

- Soot is basically unburnt fuel, its presence indicates low efficiency
- **One of the main causes of soot production is low in-chamber temperature**



NOx production

- Atmospheric air is made up of ~79% of molecular Nitrogen (N_2) and ~21% Oxygen (O_2).
- Molecular nitrogen is usually an inert gas at low temperatures.
- N_2 and O_2 , because of the **high temperatures and longer combustion times** of Diesel engines, decompose to their atomic form N and O .
- This is particularly true for Diesel engines where the **excess air** is far higher than SI engines.
- Atomic nitrogen can be oxydized and form the so-called NOx (nitrous oxydes) $\rightarrow NO$ and NO_2 .

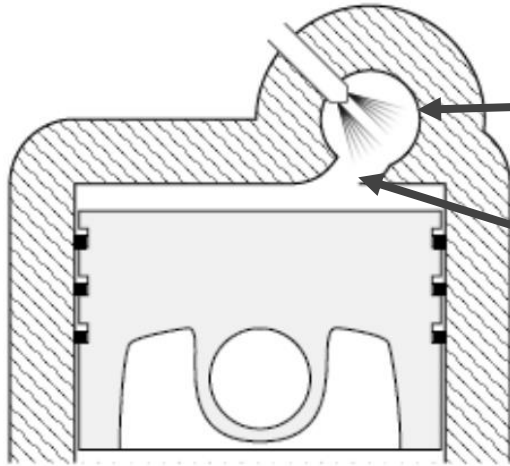
Observations

- NOx are basically due to the high efficiency of Diesel engines: high temperatures and pressures and no throttling (excess air).



Diesel engines

Indirect Injection



Combustion starts more easily and more rapidly in a small volume (initial combustion heat is more concentrated). *Reduced ignition delay.*

The shape of the prechamber and connecting hole, **create a turbulent flow** that improved the air-burning mixture mixing.

The combustion pre-chamber strongly mitigates the requirement of having very exact injection timing (a simpler fuel injection system may be used)

The combustion pre-chamber slowed down the heat release (and, subsequently the pressure) peak, reducing mechanical stresses and engine noise.

The slower heat release and larger surface exposed to the combustion heat, **reduced efficiency**

The prechamber connecting hole, induce a further pressure (and thus energy) drop that **reduces efficiency.**



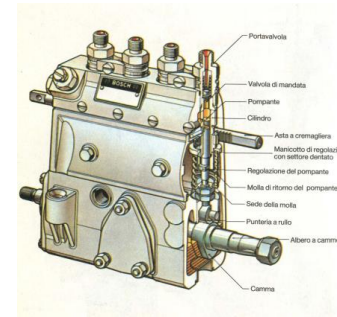
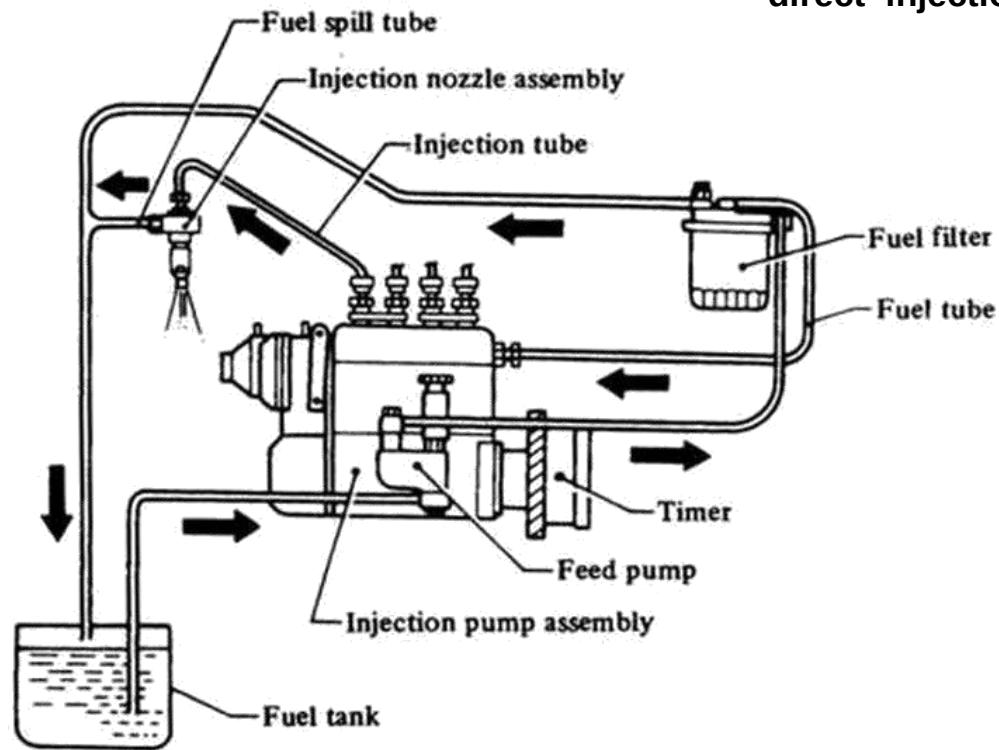
Diesel
engines

Direct Injection

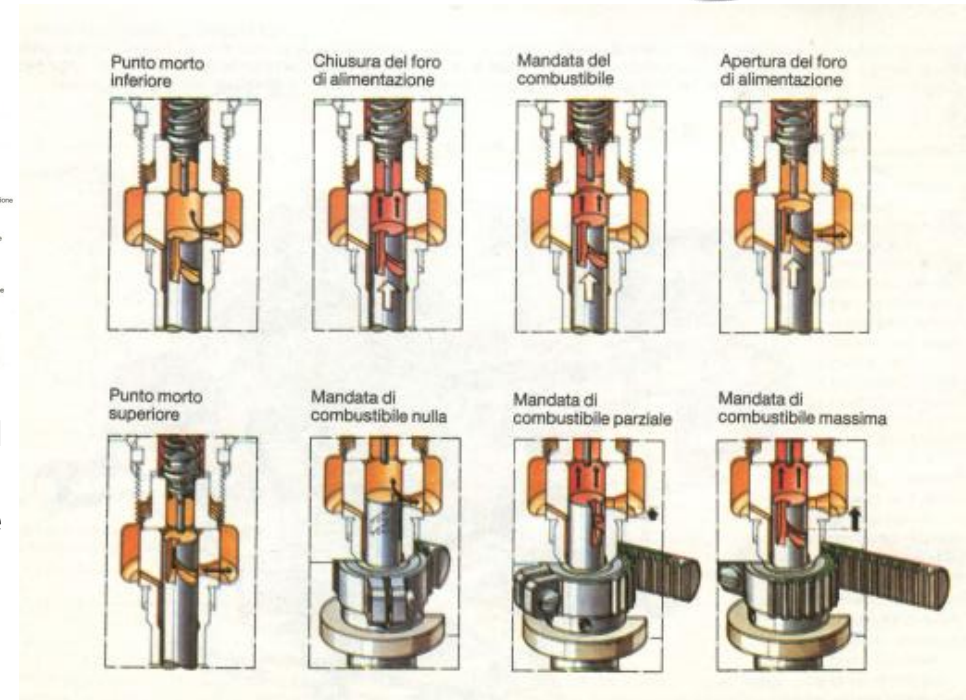
More accurate engine design to
optimize fluid dynamics
(turbulence)



The increased quality of diesel fuels and the need of
attaining higher efficiencies brought to the **adoption of
direct injection.**



The high fuel
pressure forces the
(mechanical)
injector
opening



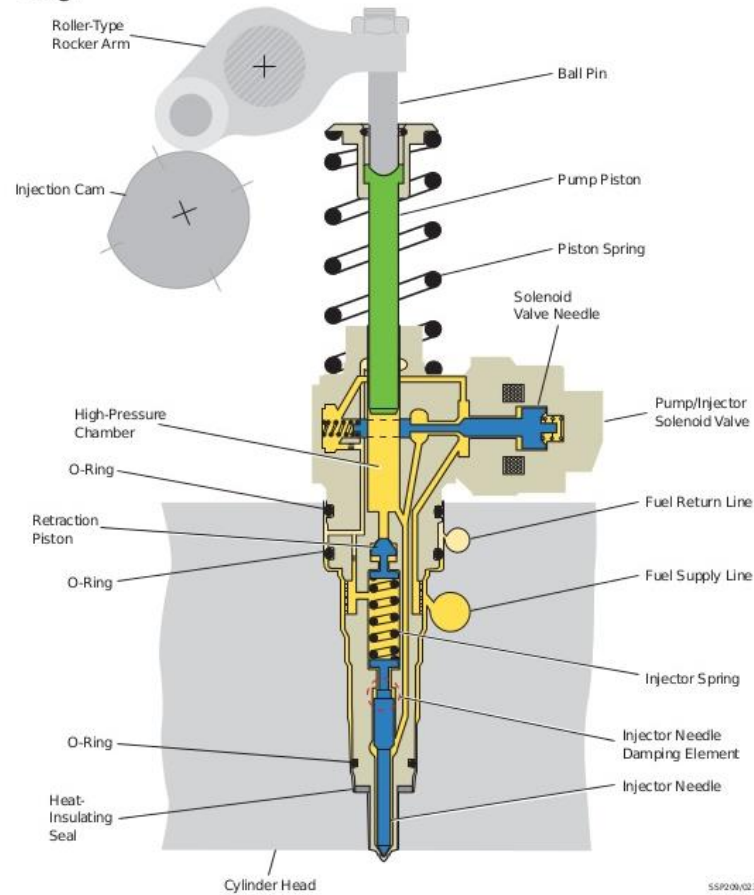


Diesel engines

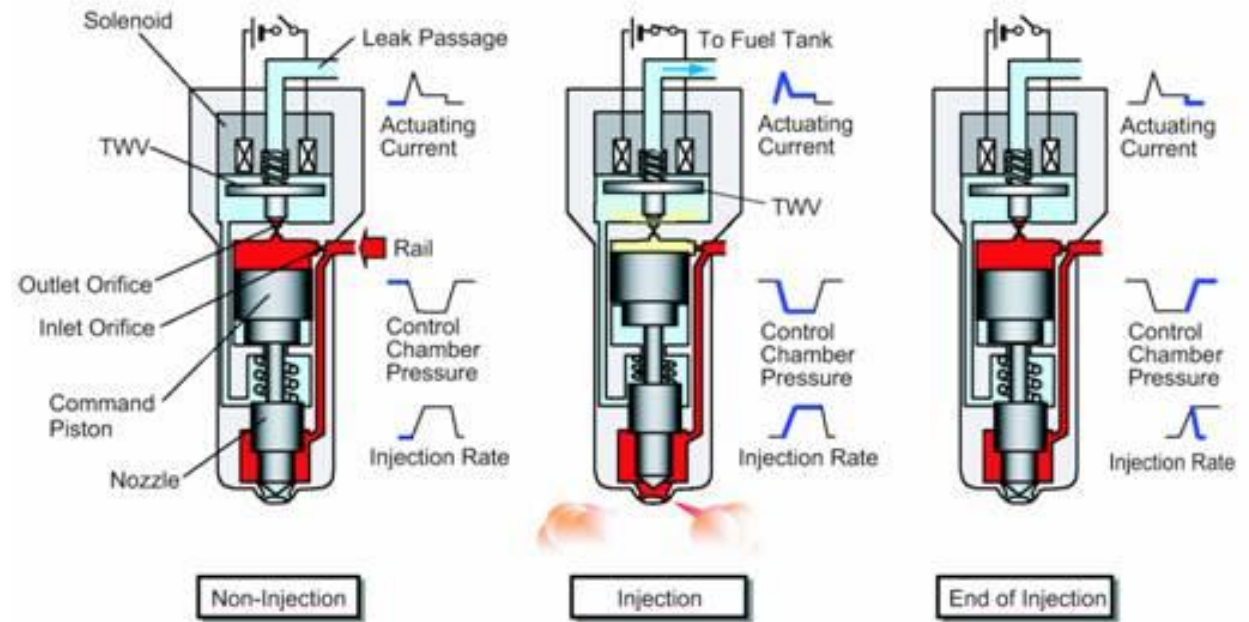
(Electronic) Direct Injection

Electronically controlled unit pump injector

Design



Electronic direct injection

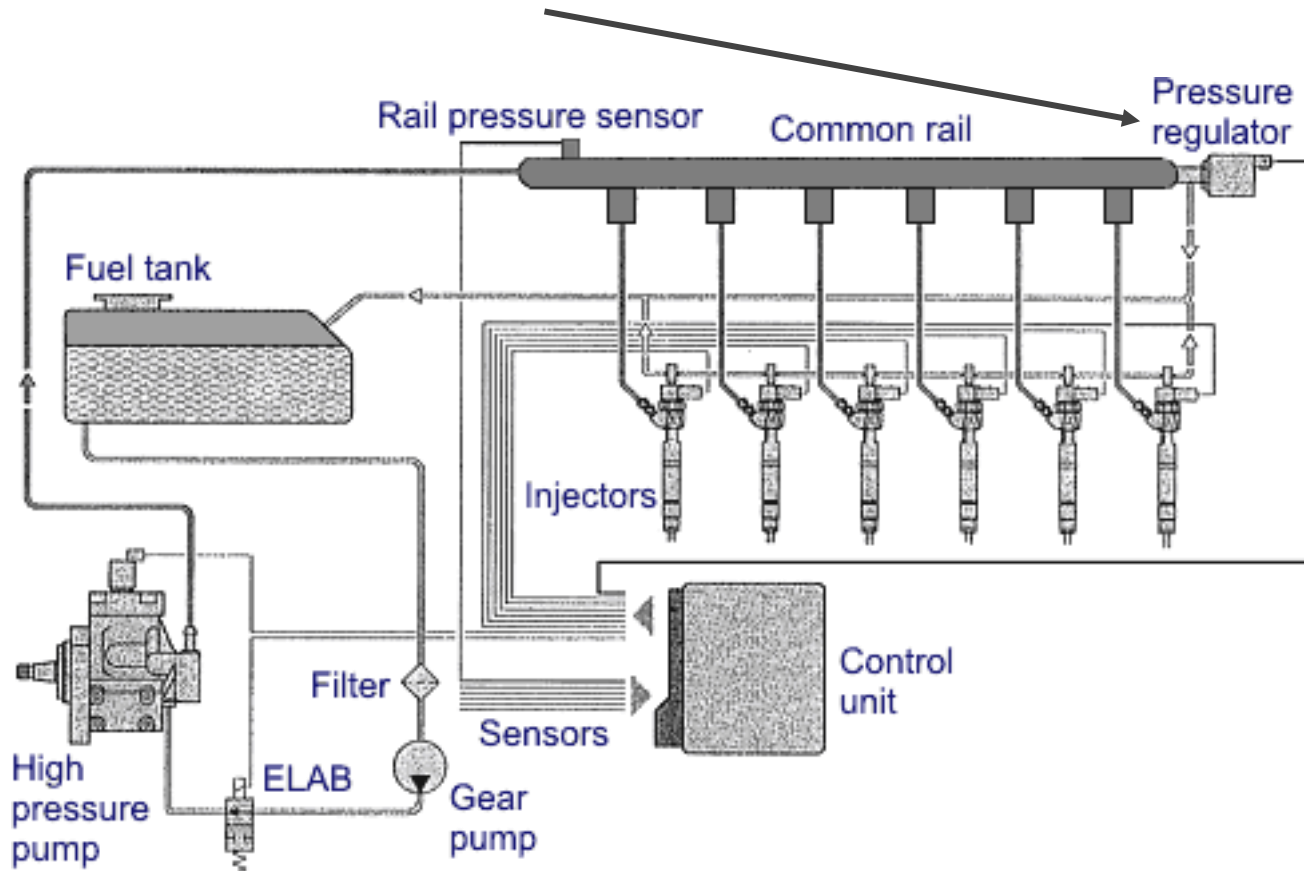




Diesel engines

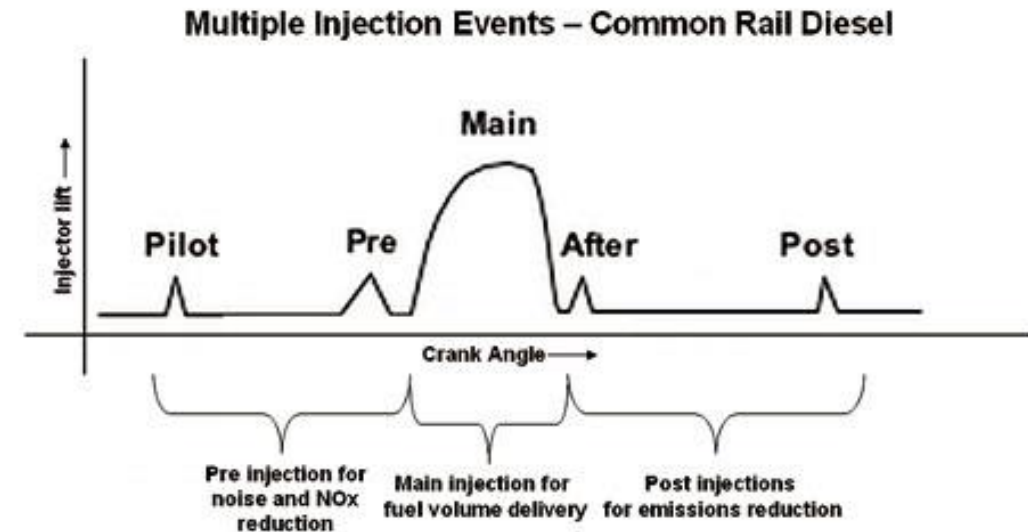
Common rail injection

The rail pressure is electronically controlled by ECM



The **common rail decouples** the available fuel pressure from the pressure provided by the injection pump, **acting as an hydraulic accumulator**.

Since fuel is always available at high pressure, an electronic injection system allows multiple injection in a very short time interval (in the order of 1-2ms)





Diesel engines

Injection phases

PILOT

Reduce ignition delay of the main injection (raises the in-chamber temperature).

The reduced ignition delay greatly improves combustion noise and reduces mechanical stresses.

PRE

Reduces oxygen availability at the time of the main injection, reduces NOx production but increases soot.

MAIN

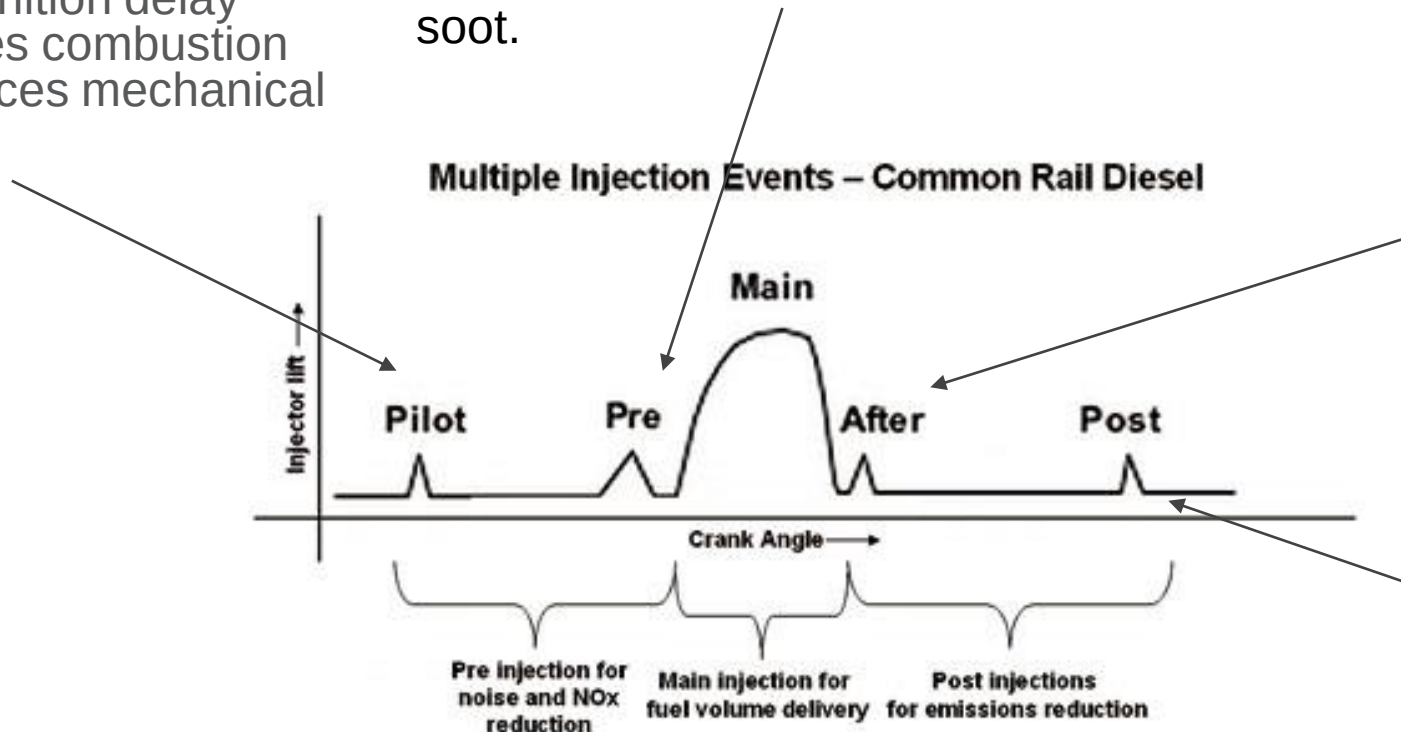
The time shape of this injection phase may be varied thanks to electronically controlled injectors

AFTER

Promotes soot oxidation by sustaining the combustion process longer in the expansion stroke

POST

This injection is not always present. It's needed for the correct operation of the aftertreatment systems (DPF).



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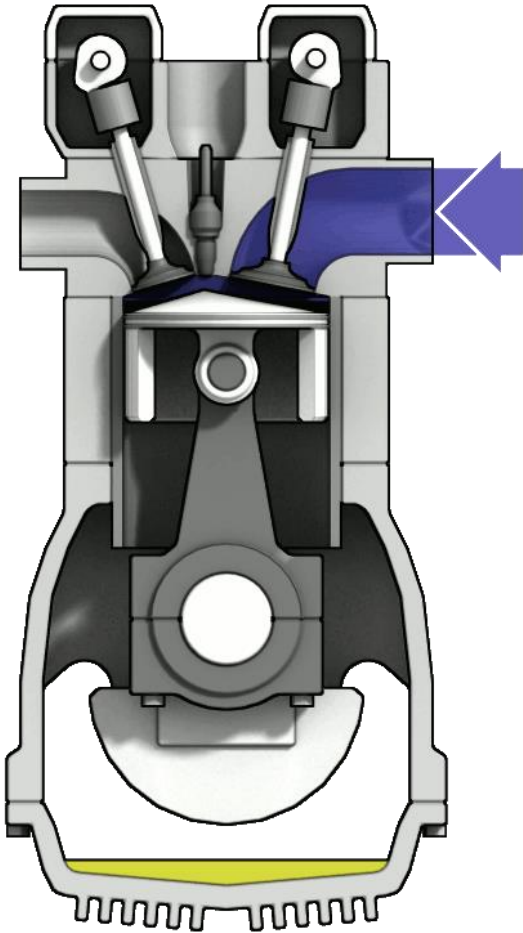
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4-stroke engine - Operation

1



1. INTAKE
2. COMPRESSION
3. EXPANSION
4. EXHAUST